Homework_2

June 23, 2026

Task 1 - Hometown Message

To assign the variable "city" I used the input() function. So what ever the user types in will be the value for "city". Then the code will print "city" along with a friendly message.

[9]:
```python
city = input('Enter the city you are from: ')
print(f'{city}, thats awesome I need to visit sometime!')
```

```
Enter the city you are from: Richmond
Richmond, thats awesome I need to visit sometime!
```

Task 2 - Break-Even

Here I created a break-even equation for a business to find out what there break-even point is.

[10]:
```python
fixed_costs = 8000 # fixed costs
unit_price = 40 # price per unit
variable_costs = 15 # costs per unit
break_even_quantity = fixed_costs / (unit_price - variable_costs) # formula
print(f'The break even quantity is {break_even_quantity}') # prints results
```

```
The break even quantity is 320.0
```

Task 3 - Operators

I completed some basic math problems first doing them by hand and then checked it with python operators.

By hand: * 25 // 4 = 6 * 25 / 4 = 6.25 * 25 % 4 = 1 * 3 + 6 * 2 = 15

[11]:
```python
print('Python Results:')
print(25 // 4) # int division
print(25 / 4) # division
print(25 % 4) # remainder
print(3 + 6 * 2) # pemdas
```

```
Python Results:
6
6.25
1
15
```

Comparison: The results that I did by hand were the same as the results python has.

Thought process: When I did the problems by hand I only used a hand held calculator to solve the problems. Then I had python print the results for the same problems I did by hand. Then I check results and they were the same.

Task 4 - Conversion

This prompts the user to type in a value for "miles" then it takes that value and puts it into the kilometer equation.

[12]:
```
miles = input('Please enter the number of miles: ')
miles = float(miles) # Converts string into a float allowing me to do math.
type(miles) # Returns as a float.
kilomters = miles * 1.60934 # formula
print(f'{miles} miles is the same as {kilomters} kilometers') # prints results
```

```
Please enter the number of miles: 26
26.0 miles is the same as 41.84284 kilometers
```

Resolving error: I knew right away I would have a small problem with this code. The variable miles would be a string not a float or an int. Because the input function returns strings. To fixed this I converted "miles" to a float (using the float() function) allowing me to perform math operators on it now allowing us to convert "miles" to "Kilometers".

[12]:
```
```